Lesson plan · for the teacher

Fair or unfair models

Data science · Level 4 · Lesson 3

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A made-up results table for two groups (or the town and village lists on the worksheet); A phone or tablet with Darsi, for the lab

Learning goals

- Find bias with error rates by group.
- Explain where bias starts.
- Plan checks before adopting a model.

Activity: Audit a model

1. Warm-up (5 min): Read the reading-app story and ask: would you have noticed the problem from the overall score?
2. Teach (8 min): Split a results table by group on the board and work out each rate.
3. Audit (15 min): Groups get results for a made-up tool, work out rates for two groups and list possible causes.
4. Plan (7 min): Each group writes three questions to ask before the school adopts the tool.
5. Wrap-up (5 min): Ask two colleagues to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- Which group carried most errors?
- What in the data could explain it?
- Who should check the results?

Exit ticket

- Group A: 3 wrong in 20. Group B: 9 wrong in 20. Write both error rates.

Try it at school tomorrow
Give groups a made-up table of a model's results for two classes. They work out the error rate for each and decide whether the model is fair.

Practice questions · for you

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1 **Where does most bias in a model start?**

◯ In the pupils ◯ In the data it learned from ◯ In the screen colour

2 **A model with high overall accuracy can still be biased against one group of pupils.**

◯ True ◯ False

3 **Each 1 in the lists is a wrong score and each 0 a right one. Finish rate(xs) so it returns the error rate as a whole-number percentage, then print the rate for each group.**

```
town = [0, 0, 1, 0, 0, 0, 0, 0, 0, 0]
village = [1, 0, 1, 0, 0, 1, 0, 0, 1, 0]

def rate(xs):
    return 0

print(rate(town), rate(village))
```

Write your program:

4 **Which steps help a school find or fix unfairness in a reading app? Choose all that apply.**

☐ Have a teacher check low scores ☐ Hide the results from teachers

☐ Ask the supplier for error rates by group ☐ Train it on more varied voices

5 **Town pupils: 6 wrong scores per 100. Village pupils: 22 per 100. What does this show?**

◯ Village pupils read worse ◯ The model is fair on average, so all is well

◯ The model is unfair to village pupils

✎ **Which tool or test at your school might work less well for some pupils? How would you find out?**

Check your answers

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1 **In the data it learned from**

A model is only as fair as its training examples.

2 **True**

A good average can hide a group with many more errors.

3

```
town = [0, 0, 1, 0, 0, 0, 0, 0, 0, 0]
village = [1, 0, 1, 0, 0, 1, 0, 0, 1, 0]

def rate(xs):
    return round(100 * sum(xs) / len(xs))

print(rate(town), rate(village))
```

The model is wrong for 10% of town pupils and 40% of village pupils: four times as many.

4 **Train it on more varied voices; Have a teacher check low scores; Ask the supplier for error rates by group**

Better data, human checks and honest reporting all help; hiding results doesn't.

5 **The model is unfair to village pupils**

The errors are the model's, and they fall far more on one group.